GEN 3.4 COMMUNICATION SERVICES

1 Responsible service

Telecommunication and Navigation Services are regulated by the South African Civil Aviation Authority, established by section 2 of the South African Civil Aviation Authority Act 1998 (Act number 40 of 1998).

Telecommunication and Navigation Services are provided by the Air Traffic and Navigation Services Company Ltd (ATNS) under authority of the Minister of Transport.

Regulating Authority:

Postal Address:

South African Civil Aviation Authority
Air Navigation Services Department: Communications, Navigation and Surveillance
Private Bag X 73,
Halfway House
1685

Telephone: +27 11 545 1000
Telefax: +27 11 545 1465
AFS Address: FAHQYAYX
Website: www.caa.co.za

Service Provider: ATNS

Postal Address:

Chief Executive Officer
Air Traffic and Navigation Services Company Ltd
Private Bag x 15
Kempton Park
1620

Telephone: +27 11 607 1383/1382
Telefax: +27 11 607 1577
AFS Address: FAATYFYX
Website: www.atns.co.za

The service is provided in accordance with the provisions contained in the following ICAO documents:

Annex 10 - Vol I & II - Aeronautical Telecommunication

Doc 8400 - Procedures for Air Navigation Services ICAO Abbreviations and Codes

Doc 8585 - Designations for Aircraft Operating Agencies, Authorities and Services

Doc 7030 - Regional Supplementary Procedures (COM Procedures for AFI)

Doc 7910 - Location Indicators

1.1 Satellite communication (SATCOM)

The following telephone numbers are to be utilised for satellite communication (SATCOM) purposes:

1) Primary number:
Johannesburg Oceanic: +27 11 928 6456

2) Secondary numbers:
Johannesburg Filter: +27 11 928 6454
Cape Town Filter: + 27 21 937 1116
King Shaka Tower: +27 32 436 5002
Bloemfontein Tower: +27 51 503 7201/6
Port Elizabeth: +27 41 51 35851

It must be emphasized that these numbers are for emergency only, when all other airborne means of communication with the appropriate ATS unit have failed.

2 Area of responsibility

Enquiries, suggestions or complaints regarding any telecommunications service within the boundaries of the Republic of South Africa should be referred to the relevant Station Telecommunication Officer or to the Director of Civil Aviation

Facilities listed in the Republics FIRs but which fall within the boundaries of Lesotho and Swaziland are provided and administered by the following respective authorities:

a) Lesotho
Department of Civil Aviation
P.O. Box 629 Maseru 100
AFS Address: FXMMYAYX
Commercial Telegraphic Address: Civilair Maseru

b) Swaziland
Civil Aviation Branch,
Ministry of Works and Communications,
P.O. Box 58,
Mbabane.
AFS Address: FDMBYAYX
Commercial Telegraphic Address:, MINWORKS MBABANE

3 Types of service

3.1 Radio Navigation Services.

The following types of radio aids to navigation are available:

DME - Distance Measuring Equipment.
ILS - Instrument Landing System.
NDB - LF/MF Non-directional Beacon
RSR - Route Surveillance Radar.
SSR - Secondary Surveillance Radar

TAR - Terminal Area Surveillance Radar.
VDF - VHF Direction-finding Station.
VOR - VHF Omni-directional Radio Range.

Radio broadcasting stations are not listed as it is considered inadvisable to use them for navigational purposes.

According to the judgement of the direction finding station, bearings are classified as follows:

Class A: Accurate to within +/-2 degrees.
Class B: Accurate to within +/-5 degrees.
Class C: Accurate to within +/-10 degrees.

Direction-finding stations have authority to refuse to give bearings or headings to steer when conditions are unsatisfactory or when bearings do not fall within the calibration limits of the station, stating the reason at the time of refusal

The use of GPS as a navigational aid has limited approval by the Civil Aviation Authority.

3.2 Mobile/fixed service

3.2.1 Mobile Service

The aeronautical stations maintain a continuous watch on their stated frequencies during the published hours of service unless otherwise notified.

An aircraft should normally communicate with the air/ground control radio station which exercises control in the area in which the aircraft is flying. Aircraft should maintain continuous watch on the appropriate frequency of the control station and should not abandon watch, except in an emergency, without informing the control radio station.

Clearance delivery service.

At JOHANNESBURG/O R Tambo International Airport a clearance delivery service is provided. This service provides:-

a) ATC clearance
b) Runway in use;
c) ATIS information;
d) Transponder code; and
e) other pertinent information.

Pilots should call on the appropriate frequency prior to start-up and pass the following information:-

i) call sign and registration;
ii) requested flight level;
iii) parking bay number;
iv) fuel endurance;
v) persons on board;
vi) any changes to filed flight plan; and
vii) any other information required by ATC.

3.2.2 Fixed Service.

Messages to be transmitted over the Aeronautical Fixed Service (AFS) are accepted only:

a) if they satisfy the requirements of Annex 10, Vol. II, Chapter 3.3.3:
b) if they are prepared in the form specified in Annex 10;
c) if the text of any individual message does not exceed 200 groups.

When it is desired that a communication exceeding 200 words be transmitted over the Aeronautical Fixed Service, such a communication shall be filed in the form of separate messages, each text of which does not exceed 200 groups.

General Aircraft Operating Agency messages are only accepted for transmission to countries which have agreed to accept Class B traffic.

AFS/SITA.

An AFS/SITA interface link is available at the O R Tambo ATC Communications Centre. In view of this, airlines entering or overflying the Republic of South Africa, are required to address their flight plans and other ATC movement messages to the following SITA address: JNBXTYF.

The O R Tambo ATC Communication Centre will ensure the onward transmission of these ATC messages via AFS within the Republic of South Africa. The above does not apply to those carriers whose flights depart from airports in the following countries: Zambia, Zimbabwe, Mozambique, South Africa, Botswana and Namibia. Neither does the above apply to those airlines in respect of which repetitive flight plans have been agreed upon.

The above is intended to reduce the difficulties currently being experienced by airlines and ATC due to ATC not receiving flight plans transmitted on AFS.

ATS DIRECT SPEECH CIRCUIT REPUBLIC OF SOUTH AFRICA/O R TAMBO INTL.

As no ATS Direct Speech Circuit currently exists between South Africa, Johannesburg ACC (Johannesburg Oceanic) and Madagascar, Antananarivo ACC and between South Africa, Johannesburg ACC (Johannesburg Oceanic) and South America, Brazilia ACC, Recife ACC (alternate) agreement has been reached to utilise the following Public Switch Telephone Network (PSTN) and fax numbers:

Antananarivo ACC Tel: 09 261 20 2233912
Brazilia ACC Tel: 09 556 1 365 1586 Fax: 09 556 1 365 1768
Recife ACC Tel: 09 558 1 462 2742 Fax: 09 558 1 462 4927
The ATS/DS and AFTN circuits between Johannesburg and Ezeiza (SAM) have now been implemented via the CAFSAT satellite network. These have been operationally tested.

3.3 Broadcasting Service

3.3.1 The FLW broadcasts are available for use of aircraft in flight:

a) meteorological broadcasts as detailed in GEN 3.5
b) automatic terminal information service (ATIS) broadcasts superimposed on the transmission from VOR stations serving certain major airports.

3.3.2 ATIS Broadcasts

ATIS broadcast are transmitted continuously throughout hours of service and are revised every 30 minutes, on the hour and 1/2 hour. Each broadcast is prefixed with an identifying letter in the phonetic alphabet, ALFA to ZULU

Broadcasts contain:

a) the latest METAR for the airport concerned;
b) information on the unserviceability of navigational, approach or landing aids; and when appropriate,
c) essential information on aerodrome conditions.

Aircraft having received the ATIS broadcasts are to inform Approach Control accordingly immediately after initial contact

3.4 Language used: English

3.5 Detailed information can be obtained in ENR 4 and AD 2

4 Requirements and conditions

The requirements and general conditions under which the communication services are available for international use, as well as the requirements for the carriage of radio equipment, are contained in the Air Navigation Regulations (ANR's) of the RSA - see sections referring to Aviation Act and Air Navigation

5 Data Link Automatic Terminal Information Service (D-ATIS). Cape Town and Johannesburg FIR

5.1 Introduction

a) There will be an Air Traffic Service (ATS) air/ground data link application service for the provision of the Data link Automatic Terminal Information Service (D-ATIS) at three of South Africa International Airports with effect from 27 October 2005.
b) Service Airports:
   1) O R Tambo International Airport
   2) Cape Town International Airport
   3) King Shaka International Airport
c) The D-ATIS data link system installed at the 3 airports have been equipped with data link capability and dedicated data link communication links have been set up with SITA AIRCOM Service to enable aircraft to access D-ATIS service via VHF data link and Satellite data link.

5.2 Area of Operation

a) D-ATIS data link service will be available to aircraft equipped with air/ground data link capability.
b) The SITA data link service uses Internet working with Cooperating Data link Service Providers (Co-DSPs).

5.3 Data Link ATIS Service Connection

a) The SITA Data link Service Designation Address of each Airport D-ATIS system is "JNBATYA"

b) The D-ATIS is available on the following VHF frequency 131.725 MHz.
c) Data link to request D-ATIS information follows the AEEC 623 and the EUROCAE ED_89A Data link Applicant System Document (DLASD).
d) The Standard Message Identifier (SMI) on the line 3 of the data link message to be used is as follows; D-ATIS: RAI (B9) - Request ATIS Report (for downlink message).
e) In the request D-ATIS report message, the following formats shall be used;
   i) Airport ID: O R Tambo International FAOR, Cape Town International FACT and King Shaka International FALE
   ii) Arrival/Departure Indicator Codes shall be as follows:
   A - Arrival ATIS (ARR ATIS)
   D - Departure ATIS (DEP ATIS)
   C- Contract ATIS (Auto Update ATIS)
   T - Terminate C mode (Terminate Auto-Update ATIS)
   E - Not Used
   iii) The Arrival ATIS and Departure ATIS are identical for each Airport.
   iv) C-mode is automatically terminated.
f) Use of voice communication

   Pilots of suitably equipped aircraft that cannot establish communication with data link ATIS should read the broadcast of the radio ATIS on the VHF frequencies.

5.4 Data Link Failure

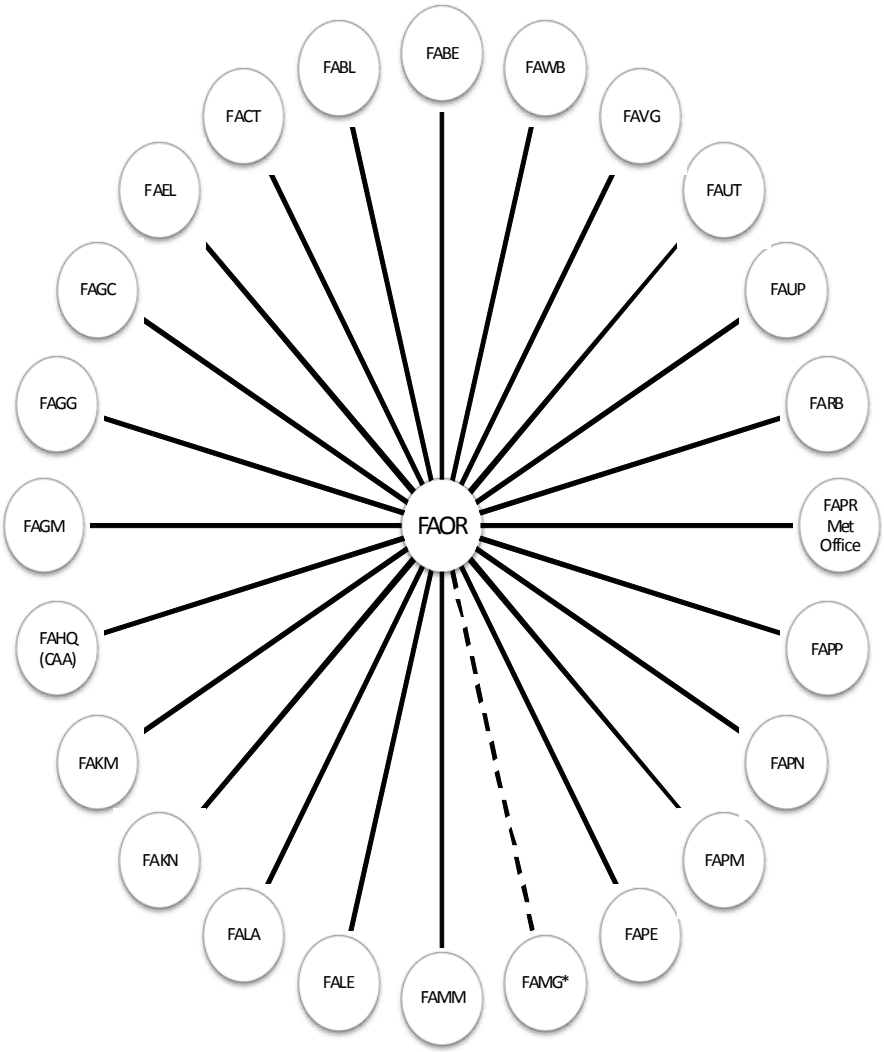
Pilots shall inform the ATC at each airport of problems encountered during the provision or alternatively call the Johannesburg technical support help line on +27 11 928 6476/7.

5.5 Hours of operation

a) FAOR: 24 Hours daily
b) FACT: 24 Hours daily
c) FALE: 24 Hours daily

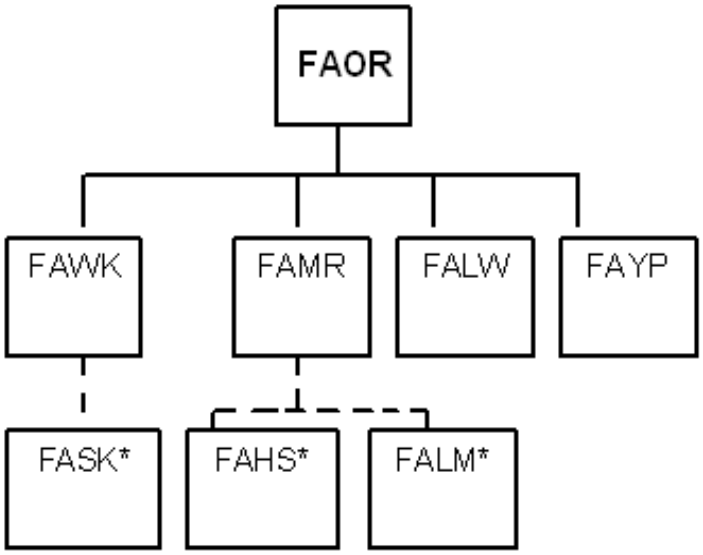
6 The South African Aeronautical Message Switching System (AMSS) is situated at the Air Traffic Control Centre at O.R. Tambo International Airport. The South African AMSS is the central Aeronautical Fixed Service (AFS) Hub for the SADC region, and therefore all AFS traffic in the region is switched through this central hub. The AMSS is monitored and maintained by the ATNS Communication Centre and is responsible for both national and International links as depicted below.

NATIONAL CIVILIAN AFTN/AMHS - SCHEMATIC



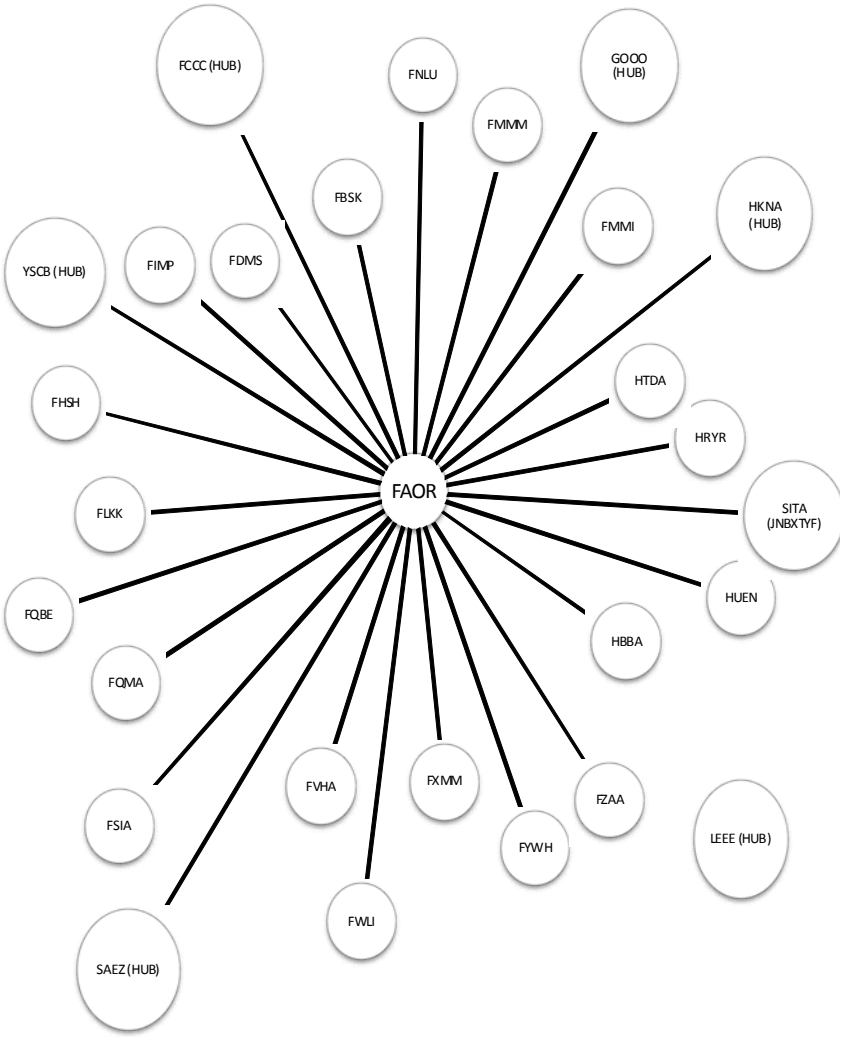
* FAMG is receiving only link utilising AFTN to e-mail

NATIONAL MILITARY AFTN - SCHEMATIC



*FASK, FAHS and FALM are connected via a dumb terminal and are receive only

REGIONAL AND INTERNATIONAL AFTN - SCHEMATIC



Aeronautical Fixed Service (Telephone)

ATS DIRECT SPEECH CIRCUITS

